

The Impact of a Clinic Patient Record System on Patient Satisfaction and Clinic Efficiency on Comfort Smile Dental Clinic

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ABSTRACT

This study delves into the challenges faced by Comfort Smile Dental Clinic as it strives to optimize its patient record system. Established in 2017 under the vision of Dr. Jim D. Bares, the clinic stands as a testament to the transformative impact of technology on the relationship between patients and doctors. The current system faces challenges in keeping up with timely updates, prompting a thorough investigation to enhance overall efficiency.

The research specifically focused on clinic patient record systems, aiming to understand their influence on both patient comfort and clinic proficiency. The ultimate objective is the implementation of an advanced Electronic Oral Health Record, with the aim of refining the existing system. The study hones in on the transition from traditional paper records to Clinic Patient Record Systems, exploring ethical considerations, societal impacts, and professional implications.

The examination places a strong emphasis on the positive outcomes expected from this transition. Anticipated benefits include improved communication, heightened patient satisfaction, and an overall increase in healthcare quality. These positive changes highlight the broader narrative of technology's pivotal role in shaping the healthcare landscape. The evolution of Comfort Smile Dental Clinic serves as a microcosm of this larger trend, illustrating how advancements in patient record systems contribute to creating a more streamlined and effective healthcare environment.

Key words: Clinical Decision Support Tools, Clinic Patient Record Systems, Data Security, Electronic Oral Health Records, Patient Portal

CHAPTER 1

1.1 BACKGROUND OF THE STUDY

The physical environment where patients and doctors meet, such as a medical clinic or hospital, has a significant impact on their interactions. Dental clinics are organizations responsible for providing dental care and treatment to patients. The development of specialized medical information systems has revolutionized medical practices, making data collection and access more efficient and faster [1]. Advances in the use of information technology are rapidly changing the way we think, reason, make decisions, and interact with others.

Comfort Smile Dental Clinic is a Dental Clinic that offers a wide range of dental services such as Orthodontic care for Brace and Invisalign, Cosmetic dentistry that is for whitening, veneers, and crowns, Preventive care that includes cleanings, exams, and fluoride treatments, and other services that is related to Dental Services. Comfort Smile Dental Clinic is committed to providing patients with the highest quality dental care in a comfortable and relaxed setting. The clinic's team of experienced and compassionate dentists and hygienists are dedicated to helping patients achieve their dental goals.

Their company was first established in 2017 and located at New Pandan, Panabo City, and it was operated and managed by Dr. Jim D. Bares, a licensed Dentist. Their first clinic was located on Daang Maharlika Highway, Panabo City for a year and a half. Then they transfer near Ybañez Subdivision and Panabo Wharf Road, Panabo City for almost half a year as well. After transferring with two clinics, they decided to change location at New Pandan, Panabo City at year 2021 for good. Their organization aimed to meet customers' needs, enhance patient care, and promote wellness to society.

Comfort Smile Dental Clinic is open Monday to Friday from exactly 9:00 am until 5:00 pm. On weekends, they start operating at 1:00 pm until their 6:00 pm. A Secretary is assigned and tasked to ensure that patient shall be cater by the Doctor, advising and answering patients' queries and scheduling an appointment. Their Dental clinic is using a Clinic Patient Record System to store, collect, and process patient's information. The clinic's team of experienced and compassionate dentists are dedicated to helping patients achieve their dental goals.

1.2 STATEMENT OF THE PROBLEM

The Comfort Smile Dental Clinic is currently evaluating the sustainability and effectiveness of its patient record system. The clinic's efficiency in utilizing resources to achieve its goals and address patient needs, following the guidelines set by CMS in 2022. Patient satisfaction matters because it can lead to improved patient outcomes [2]. The current patient record system at Comfort Smile Dental Clinic struggles with efficient updates, hindering the accurate and timely representation of patients' medical conditions and treatment plans. This can lead to potential delays in medical decision-making, compromising the overall quality of patient care. Additionally, it contributes to data inconsistencies, which is critical for ensuring that patient records are reliable, up-to-date, and accurate. The absence of dental healthcare features also hinders the system's ability to provide specialized tools and

functionalities essential for dental care. Upgrading the system is important for enhancing efficiency, accuracy, and overall quality.

To achieve this, a seamless patient record system and efficient management of clinical functions and administrative tasks are crucial. Dental assistants generally explained that there's more positive effects on communication with storing, collecting, processing patient's information in Clinic Patient Record System.

1.3 OBJECTIVES OF THE STUDY

- To develop a comprehensive understanding of clinic patient record systems
 - Encompassing key components such as data entry, retrieval, security protocols, and interoperability, to facilitate effective communication and collaboration within healthcare settings.
- To evaluate the impact of implementing the clinic patient record system on patient's comfort and clinic proficiency.
- To establish advanced Electronic Oral Health Record for accuracy and accessibility of patients' oral health care management.

1.4 LITERATURE REVIEW

Storing necessary information about a patient's health status and updating patient's status is an important aspect of provider-patient communication and doing so ultimately empowers patients. Although providers were mostly pleased with their current system, they identified some usability-related issues as performance obstacles occurred [2]. The availability of a system which can help the Dentist to manage their patient record will satisfy the needs of the patients. The Clinic Patient Record System (CPRS) will enable users to access patient records, adding and updating patient records stored in the database.

Dentists in Myanmar keep only basic paper-based dental records, which are often incomplete and inaccurate as patients often do not bring them to subsequent visits. This can lead to errors in diagnosis and treatment and weaken the quality of patient care. The Clinic Patient Record System would be a better solution, as they would be more accurate, complete, and accessible to dentists. CPRS would also make it easier to share patient records with other healthcare professionals, which could improve the coordination of care. Implementation of a new electronic data-entry system in their dental clinics has been challenging for most dentists for several reasons, such as financial barriers, human resources barriers, knowledge barriers, and legal and regulatory barriers. In addition, even after the implementation of CPRS in dental clinics, paper-based patient records might still be useful to CPRS in any case of loss of information in the system [3]. This concludes by recommending that the first step towards implementing CPRS in Myanmar should be to improve the content and structure of paper-based dental records. This could be done

by developing national guidelines for dental records, and by providing training for employees on how to keep accurate and complete records. The Clinic Patient Record System (CPRS) is a paperless system that aims to facilitate patients' record documentation and simplify the transfer of information between clinical practices. It is believed that CPRS will improve patient care by providing more comprehensive and accurate information on the medical and dental history of an individual and making it easier to share this information between practices. Empowering patients to improve their satisfaction and customer loyalty is now considered an important part of healthcare. One way to do this is to inform patients and share the necessary information on their health status and available treatment options with them by using the CPRS [4]. With this, they can easily access information without searching manually. CPRS can help the dentist in terms of the time spent looking for certain patient information.

Electronic Oral Health Records (EOHRs) are digital repositories of patient health information that can be accessed and updated by authorized healthcare providers [5]. They serve as a comprehensive record of an individual's medical history, with various aspects of their health. They also facilitate the sharing of health information across different healthcare settings, ensuring that all providers involved in a patient's care have access to their complete medical history. From this system, this have the similarity to the Clinic Patient Record System that may include fields for the information that is essential for providing patient care, such as the patient's medical history, medications, and allergies. This can save clinicians a significant amount of time and effort. CPRS is a powerful tool that can help clinicians to be more productive and efficient.

By reducing the time that clinicians spend on administrative tasks, CPRS can free them up to focus on providing patient care which results on patient satisfaction [6]. Clinic Patient Record Systems is an essential tool for dental clinics of all sizes. By implementing a well-designed record system, dental clinics can improve the quality of patient care, increase efficiency and productivity, reduce costs, improve communication and collaboration, and enhance patient satisfaction. A more advanced Clinic Patient Record System has been proven that it is the ultimate solution to this problem.

The implementation of a comprehensive dental software system could revolutionize the way dental clinics operate, leading to enhanced efficiency, improved patient care, and streamlined administrative processes. By automating various tasks that are currently performed manually, the system could free up dental staff to dedicate more time to providing direct patient care, enhancing the overall quality of service.

As Maguire, from St. Paul University of the Philippines, stated that, one of the key benefits of the system lies in its ability to manage patient appointments effectively. The scheduling feature would enable seamless appointment booking, and tracking up-to-date appointments, significantly reducing the number of disarranged appointments. This would not only optimize the clinic's schedule but also free up staff time that would otherwise be spent on handling missed appointments. Additionally, the system's ability to store and manage patient records securely would provide easy access to critical medical and dental

history, treatment plans, and other relevant information, enhancing the quality of care provided to patients.

The dental software system offers an abundance of benefits that would transform the operations of dental clinics. By automating various tasks, enhancing patient engagement, and streamlining administrative processes, the system would not only improve efficiency and productivity but also contribute to a higher standard of patient care. As computerization continues to permeate the healthcare landscape, it is important to adapt technological solutions to the evolving needs of end-users, program managers, and clinicians. This ensures that the system remains effective, efficient, and aligned with the specific requirements of dental clinics. [8]. These computer-based tools provide clinicians with comprehensive and accurate patient data [9]. Also known as computer-based patient records (CPRs) [10], EPRs represent the next step in the digitization of dental practices. Their benefits, particularly for large clinics, are evident. EPRs facilitate document storage, access, and record control. To safeguard sensitive patient information, it is essential to encrypt all data stored within electronic health information systems. [11]. With the help of security measurements and safety, this could help the dental clinic for protecting patient privacy, preventing identity theft and financial fraud, maintaining compliance with regulations, and preserving the reputation of healthcare organizations. It is a critical component of data security in the healthcare industry, ensuring that sensitive patient information remains safe in the global wide aspect of technology.

When you use a Clinic Patient Record System (CPRS), you don't have to worry about losing your paperwork. When you come back to the clinic, the dentist can access your information online, so you don't have to fill out any forms. This saves time and hassle-free appointments. Patients will also appreciate the fact that the clinic uses a fast and reliable system, which shows that they care about their patients' safety and convenience. There is no need for a safe, large repository of papers since data is in electronic form which is more secure. The search for patient information is made easier since data is stored orderly and in one place. More importantly, data can be easily accessed via the web. In 2007, a comprehensive review of dental patient satisfaction studies revealed that the common dimensions or factors assessed in patient-satisfaction surveys were primarily concerned with the perceived skills of the dental practitioner, interpersonal aspects of care, convenience of scheduling and accessibility, financial considerations, and the overall clinical environment [16]. These dimensions represent the key aspects of the dental care experience that influence patient satisfaction.

A Dental Clinic Information System studied in University of Mindanao, Davao City [12], is used to track patients' personal and health information continuously. As a result of the development of the system, clinicians can now see new opportunities for storing, collecting, and manipulating patient's information in a way that can be more practical for their daily tasks. Clinics gain a competitive advantage over those still using conventional systems. Dental clinics must have a reliable method of managing business processes, securing patient records, and scheduling patient appointments.

The productivity of practitioners could be enhanced in at least three ways. Reducing the time needed to find missing records or to wait for records already in use.

CPRs make it easy for clinicians to find and access patient records quickly and easily. This can save a significant amount of time, especially in busy clinical settings. Additionally, CPRs allow multiple clinicians to access the same patient record simultaneously, which can eliminate the need to wait for records to be released by other clinicians. Reducing, if not eliminating, the need for redundant data entry. CPRs can eliminate the need for redundant data entry by automatically populating fields with information that is already stored in the system. When a clinician creates a new patient record, the system can automatically populate fields with the patient's name, address, date of birth, and insurance information. This can save clinicians a significant amount of time and effort. Reducing the time needed to enter or review data in records that have been streamlined to eliminate unnecessary information. CPRs can be used to streamline patient records by eliminating unnecessary information. This can make it easier and faster for clinicians to enter and review data.

CHAPTER II

2.1 PROPOSED METHOD

The study will employ qualitative methods to gain a deeper understanding of the impact of the new Clinic Patient Record System on patient satisfaction and clinic efficiency. Qualitative data will be collected through semi-structured interviews with patients and staff members [13]. Qualitative research is a methodological approach that delves into and provides in-depth understanding of real-world phenomena. This research facilitates hypothesis generation and further exploration and comprehension. Qualitative research methods gather and analyze participants' experiences, perceptions, and behaviors to gain insights into their perspectives and experiences.

While many people emphasize interpretation in qualitative research, it is also an important aspect of quantitative research. For instance, the results of a regression analysis and the form of meta-analysis provided by factor analysis both require interpretation. However, there is no interpretation of raw quantitative data, such as the numbers in a table.

A common element in qualitative research is that researchers must thoroughly engage with their data to gain a deep understanding of the subject matter [14], regardless of the type of empirical material being analyzed. This observation is linked to the fact that qualitative researchers frequently adjust their focus and research design as their studies progress, often up until the very end of the project. By using this method, the study will provide a comprehensive understanding of the impact of the new clinic patient record system on patient satisfaction and clinic efficiency. Qualitative assistance aims to provide detailed insights into the situation, the decisions behind its implementation, and how it influences clinic efficiency and patient satisfaction. The findings of the study will be used to inform future decisions about the implementation and use of the new system.

2.2 PROPOSED PROCEDURES

In the pursuit of knowledge and the desire to make a positive impact on society, a team of researchers embarked on a mission to find an organization that aligned with their values and offered opportunities for meaningful collaboration. Their search led them to Comfort Smile Dental Clinic, a home of care and compassion within the community. They then introduce themselves and explain the purpose of their visit. They were met with a warm smile and genuine interest from the dental assistant, who eagerly provided insights into the clinic's system and its capabilities. The researchers asked permission to conduct a study that would explore and help elaborate the clinic's purpose as well as the system they're using. Recognizing the potential benefits of this endeavor, the clinician readily granted. To gather comprehensive information, the researchers scheduled a visit for a thorough interview of the clinic's system and its processes. Before proceeding with an interview, the team must first obtain letter of approval from the dean, BSIS chairperson, and instructor, and subsequently submit the signed letter to the dentist for signing the letter.

Through in-depth interviews, the researchers aimed to capture the essence of Comfort Smile Dental Clinic's advancement and approach to providing exceptional dental care with the use of Clinic Patient Record System. They diligently documented significant data during these discussions using voice recording and a photo to capture. They sought to understand its impact in the development of patient satisfaction and clinic efficiency with its commitment to patient well-being, its dedication to employee satisfaction, and its contributions to the community's health and well-being. This comprehensive approach aimed to shed light on the multifaceted effects of the system on dental practitioners.

Researchers will contribute to uncovering the secrets behind its success and identifying areas for potential growth and optimization. Through this, they hoped to make a meaningful contribution to the clinic, its employees, and the community it served.

2.3 RESEARCH LOCALE AND RESPONDENTS

The study was conducted at the Comfort Smile Dental Clinic, located at New Pandan, Panabo City, Davao del Norte. The research focused on examining the impact of the clinic's patient record system on patient satisfaction and clinic efficiency, gathering data from both patients and clinic assistant. The research focused on scrutinizing the daily operations and patient interactions within the clinic. To ensure a comprehensive analysis, the participants selected for the study represented a diverse group, encompassing dental assistant, and the dentist. This broad spectrum of individuals was chosen deliberately to capture a range of perspectives on the patient record system. By including the viewpoints of both those providing care and those managing administrative aspects, the study aimed to gain insights into the system's influence on clinic efficiency and patient satisfaction.

Through this varied selection, the research sought to explore the differences to perceive and experience the patient record system's functionalities. The focus on diverse perspectives ensured a more holistic understanding of the system's impact, encompassing both clinical and administrative dimensions. With the help of this diverse group of

participants, the study hoped to offer insightful analysis that captured the complexity of the clinic's operations and the function of the patient record system within them

CHAPTER III

3.1 RESULTS AND DISCUSSION

According to the researcher's interview and observation, the impact of the Clinic Patient Record System in the organization holds importance in different aspects. Health professionals, such as medical and dental clinicians, have scant understanding of patients' experiences and perceptions of satisfaction, affecting both the efficiency of operations and the professional dimension of healthcare.

A well-functioning CPRS can significantly enhance in terms of professional aspect [16] in patient satisfaction by streamlining processes, improving communication, and facilitating patient involvement in their care. CPRS enables efficient appointment scheduling and patient flow management, minimizing wait times and reducing patient frustration. This also facilitates clear communication between healthcare providers and patients, allowing patients to access their records, understand their diagnoses, and actively participate in their care decisions. These technologies can elevate the quality of patient care through instantaneous access to patient histories, medications, and treatment plans, facilitating well-informed decision-making. Safeguarding sensitive patient information reinforces adherence to professional ethics and standards [17]. Continuous learning opportunities offered by advancing IT/IS tools and technology benefit healthcare professionals. The incorporation of IT/IS into a Clinic Patient Record System not only improves operational efficiency but also has a positive impact on the professional aspects of healthcare by emphasizing data protection and promoting ongoing learning.

Studies by Shabbir and Asif et al. [18] have demonstrated a strong link between patient satisfaction and the services the health clinicians offer in social aspect. Effective public education initiatives can foster trust in healthcare providers and positively impact hospital administration. The implementation of CPRS in dental clinics introduces a myriad of social aspects, it fosters better doctor-patient relationships that may access to their own healthcare records, fostering a sense in their healthcare decisions. As healthcare providers can access comprehensive patient histories, enabling personalized and informed care. It may also contribute to a more patient-centric experience and a positive social impact through increased engagement and improved communication positively impacting the clinic's reputation within the community. Clinic Patient Record System is the first step to transforming health care into efficient and productive. This helps sustain better health care by improving all aspects of patient care, including safety, effectiveness, patient-centeredness, communication, education, timeliness, efficiency, and equity [19]. Improved efficiencies and coordination of health care services, as well as by reducing the use of paper forms and providing convenient access to medical records and reviewing diagnoses from time to time.

From an ethical standpoint, Clinic Patient Record System at Comfort Smile Dental Clinic can positively

impact ethical aspects by ensuring ethical consideration in healthcare patient confidentiality. Data integrity for accuracy, consistency and reliability of the information stored in a clinic. Efficient access to medical records ensures that authorized healthcare professionals can quickly retrieve and review patient information when needed. It promotes ethical standards by safeguarding sensitive information, enhancing trust between patients and the clinic, and adhering to privacy regulations.

Physicians must cultivate trustworthiness to establish strong patient-physician relationships [20]. Trustworthiness is especially important when patients, during their most vulnerable moments, entrust their well-being to healthcare providers. Integrity, encompassing the harmonious alignment of emotions, knowledge, aspirations, and moral values, is another essential virtue for physicians. Both professional integrity and personal integrity are necessary for physicians to uphold high ethical standards in all aspects of their practice, as professional integrity alone may not encompass all situations. Safeguarding patient data is a fundamental obligation of healthcare professionals, and strict regulations govern the handling of sensitive patient health information [21]. The benefits of EHR systems in healthcare hinge on high-quality data to transform healthcare delivery, enhance patient safety, promote patient engagement, and optimize healthcare cost reimbursement. These factors directly influence ethical behavior in the workplace.

CHAPTER IV

4.1 CONCLUSION

In conclusion, the review of related literature emphasizes the pivotal role of Comprehensive Patient Record Systems (CPRS) in addressing the challenges associated with paper-based records. The research underscores that CPRS, by efficiently storing, collecting, updating, and deleting information, emerges as the ultimate solution for dental clinics. Updating the system and creating a history section in it will have an impact regarding the clinician's insight.

The findings from the literature also illuminate the specific reasons behind the preference for CPRS among dentists, citing the stark differences between paper-form records and system-based records. System-based records have been subjected to be more efficient, creates productivity, and effective. These differences, as highlighted in the study, underscore the necessity and advantages of transitioning to CPRS.

4.2 RECOMMENDATION

Exploring the current state of information systems within dental clinics and delving into potential refinements and advancements in Clinic Patient Record Systems (CPRS) holds substantial promise for pushing the boundaries of knowledge and technology in the realm of dental care. By scrutinizing the existing landscape of information systems, researchers can identify areas for enhancement within the system and the organization itself.

- **Establish a secure patient portal:** Comfort Smile Dental Clinic should develop a secure patient

portal that allows patients to access their medical records, receive information about potential treatments, and communicate with healthcare providers.

- **Safeguard patient data with encryption:** Leverage health information exchanges with a secure and controlled data sharing of sensitive and personal data at rest and in transit securing it against potential data breaches, ensuring seamless coordination of care which only Authorized Personnel can only access and share the information.
- **Enforce data safety measures:** Regularly audit and monitor system access and data usage to identify and address potential security vulnerabilities, maintaining data integrity and patient privacy.
- **Integrate clinical decision support tools:** Embed evidence-based clinical decision support tools into the CPRS system, including clinical guidelines, patient data analysis, personalized risk assessments, and treatment recommendations.
- **Implement real-time alerts:** Implement real-time alerts and reminders to inform clinicians of potential medication interactions, adverse drug events, or other safety concerns, enhancing patient safety.

This inquiry may encompass evaluating user experience, system integration capabilities, and the adaptability of CPRS to evolving technological standards. Furthermore, a thorough investigation into potential improvements can shed light on innovative features or functionalities that could elevate the overall quality of dental care services. The findings may serve as a valuable resource for practitioners, policymakers, and technology developers, fostering a collective understanding of how advancements in information systems can positively impact patient care, clinic efficiency, and the integration of evolving technologies in the ever-evolving landscape of dental healthcare.

ACKNOWLEDGEMENT

The completion of this study on the impact of a clinic patient record system on patient satisfaction and clinic efficiency at Comfort Smile Dental Clinic would not have been possible without the invaluable contributions of several individuals and organizations.

First and foremost, I would like to express my deepest gratitude to the Almighty, His grace and compassion have been my constant source of strength and inspiration.

We would also like to express our deepest gratitude to the Comfort Smile Dental Clinic's management for their unwavering support and cooperation throughout the study. Their willingness to open their doors to us and provide us with access to their patient records and clinic operations was instrumental in enabling us to conduct this research. Especially, the Dental Assistant of Comfort Smile Dental Clinic, Mrs. Irene Guardie, who graciously agreed to share

their time and experience with us. Her willingness to participate was essential to the success of this research.

To our faculty instructor, Mr. Philip John L. Paja, for his exceptional guidance and keeping us up throughout this course. His passion for the subject matter was infectious, and the ability to explain complex concepts in a clear and engaging manner made learning enjoyable and rewarding. His impact on our learning experience has been profound, and we are all truly grateful for that,

Finally, to thank our families and friends for their unwavering support during this study. Their encouragement and understanding were invaluable to us.

We are proud to present the findings of this study and hope that they will contribute to the improvement of patient care and clinic efficiency in dental settings around the world.

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Documentation



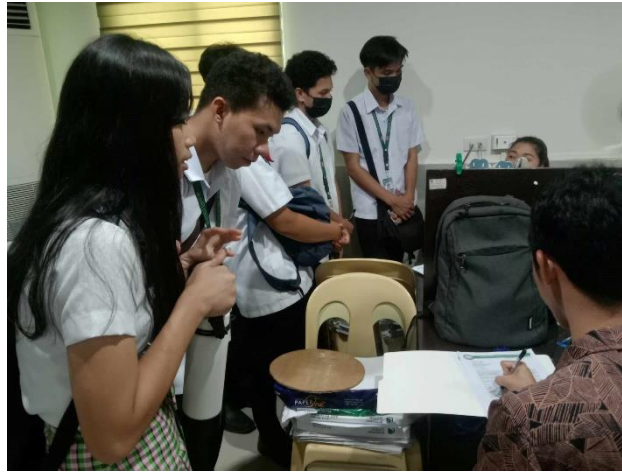
Picture 1

October 27, 2023, at Comfort Smile Dental Clinic located in New Pandan, Panabo City with Mrs. Irene Guarde asking for their system, as well as asking of approval to conduct a study in their organization.



Picture 2

November 7, 2023, Dr. Hazel Gameng, the BSIS Program Chairperson, signs the letter of approval, while on the right, Mr. Philip John Paja, a faculty instructor, does the same.



Picture 3

November 7, 2023, Philip John L. Paja, a faculty instructor, signing the letter of approval.



Picture 4

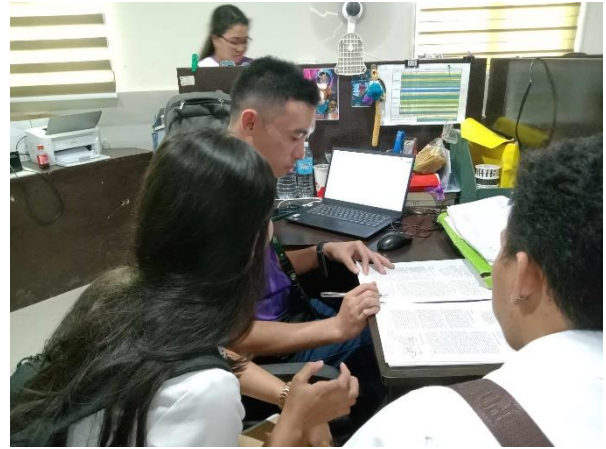


Picture 5

November 8, 2023, at Comfort Smile Dental Clinic with Mrs. Irene Guardie conducting an interview with her in regards with their organizations, and their system.



Picture 6



Picture 7

A photo of Sir Philip John L. Paja, the faculty instructor, reviewing the manuscript.



Picture 8



Picture 9

Documentation photo of checking of the revisions and important parts of the manuscript.

Curriculum Vitae

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